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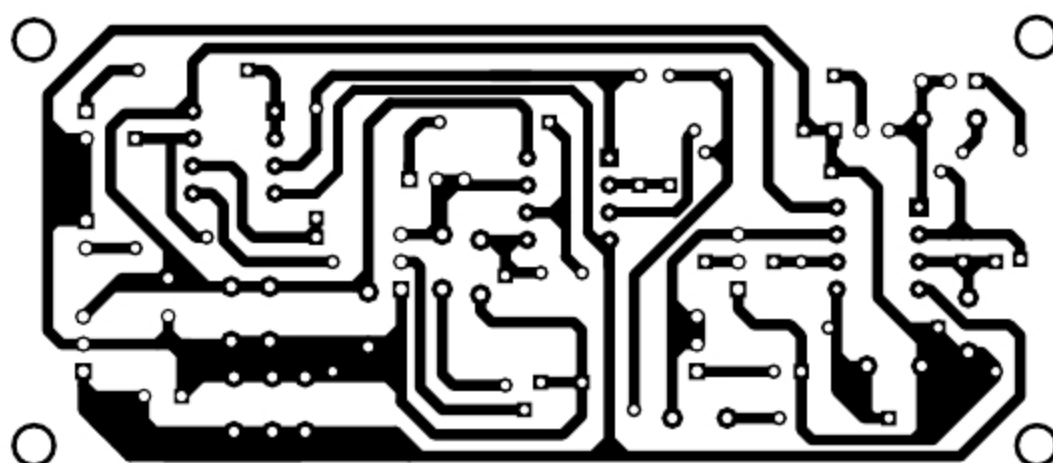


Fig. 2: Actual-size PCB layout for triple sine wave generator

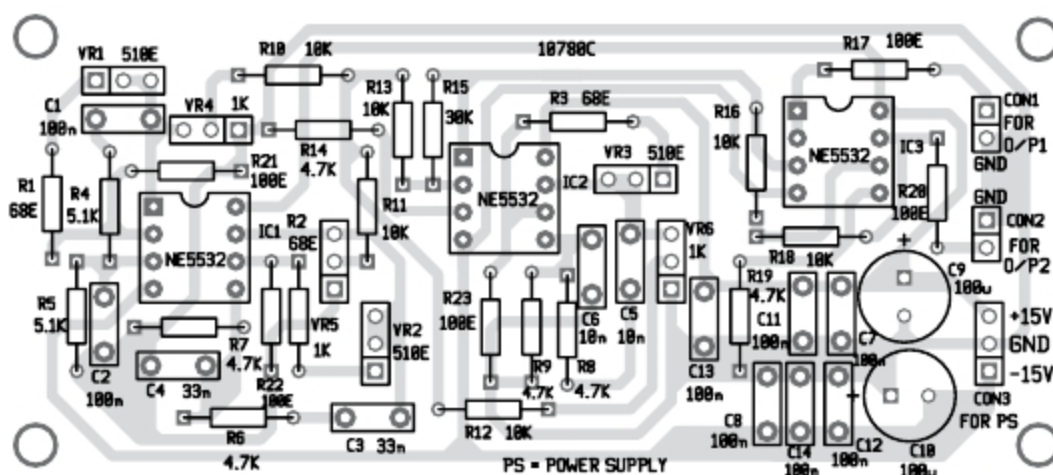


Fig. 3: Components layout of the PCB

the first and second oscillators, respectively, while the output of the third oscillator is obtained from pin 7 of IC2.

All the three signals produced by these three oscillators are mixed at IC2. Pin 1 through pin 3 of IC2 and associated components operate as a signal mixer.

The amplitudes of input signals at pin 2 of IC2 are adjusted with potentiometers VR4, VR5, and VR6.

The triple sine wave generator has two outputs. The forward output is buffered by first op-amp formed by pin 1 through pin 3 of IC3, and the inverted output is buffered with second op-amp formed by pin 5 through pin 7 of IC3. Sine wave output (O/P1) is available across CON1, and its inverted output (O/P2) is available across CON2.

The triple sine wave generator is operational in $\pm 5V$ to $\pm 15V$ range, but it is better to use it with power supply in the range of $\pm 7V$ to $\pm 15V$ only.

Construction and testing

An actual-size PCB layout for the triple sine wave generator is shown in

Fig. 2 and its components layout in Fig. 3. Assemble the circuit on the PCB. Connect $\pm 15V$ dual power supply across CON3, and your circuit is ready to use.

The given circuit can produce three sine-wave signals (312Hz, 1026Hz, and 3400Hz) simultaneously, as listed in the table. These outputs are available at CON1 and the inverted outputs at CON2 connector.

The circuit needs a simple adjustment of the amplitude for each oscillator. The potentiometers VR1, VR2, and VR3 are used to adjust the output amplitude of the oscillators around IC1 and IC2. We can replace these potentiometers with appropriate fixed resistors. **EFY**

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